Large Language Models for Generative Recommendation and User Behavior Modeling

Abstract

Recommender systems function as the foundational infrastructure of the modern digital information ecosystem, actively mitigating information overload by aligning user preferences with relevant content. Over the past decade, the dominant paradigm has relied heavily on multi-stage discriminative pipelines—typically comprising separate retrieval, ranking, and reranking modules—powered by collaborative filtering and deep neural networks. However, the emergence of Large Language Models (LLMs) and advanced generative architectures has catalyzed a fundamental paradigm shift toward generative recommendation. By reformulating the recommendation problem from a discriminative scoring procedure into an autoregressive sequence generation task, generative models directly synthesize target items or item identifiers from the entire corpus, bypassing the structural bottlenecks of traditional multi-stage pipelines.

This manuscript systematically reviews the underlying mechanics of this transformation, organizing the rapidly expanding literature into coherent method families. We analyze the evolution of Recommendation as Language Processing (RLP) and critically examine innovations in semantic tokenization, including Residual-Quantized Variational AutoEncoders (RQ-VAE), soft-routing mechanisms, and cascaded sparse-dense representations. Furthermore, we explore sophisticated methodologies designed to fuse structural collaborative filtering signals with the semantic reasoning spaces of LLMs, overcoming the inherent collaborative blindness of pure text models. In the context of user behavior modeling, we investigate the profound challenges of encoding lifelong interaction sequences, detailing breakthroughs in recurrent preference memory, multi-stream decoders, and synthetic behavior generation. Finally, we evaluate advanced preference alignment techniques—such as Group Relative Policy Optimization (GRPO) and progressive preference adaptation—alongside critical real-world challenges related to industrial scalability, temporal generalization, machine unlearning, and algorithmic fairness.

Introduction

In the contemporary era of information explosion, the primary objective of information science remains the accurate matching of user intent with relevant digital content[1]. Historically, industrial recommender systems have been engineered using multi-stage cascading architectures. These traditional pipelines explicitly separate the recommendation process into candidate generation (retrieval), precision scoring (ranking), and policy-based re-ranking[4]. While deep learning-based collaborative filtering models, such as two-tower networks and deep learning recommendation models, have achieved remarkable industrial success, they suffer from inherent architectural limitations. These include semantic gaps between isolated retrieval and ranking modules, feature fragmentation, stage inconsistency, and a pervasive inability to generalize to cold-start scenarios or perform zero-shot cross-domain transfers[5]. Furthermore, conventional discriminative models typically operate on static embeddings derived from short-term behavior, severely lacking the ability to reason over high-level user intents or process open-ended, natural language queries[5].

The advent of Large Language Models (LLMs) has introduced a transformative alternative. Pre-trained on massive multi-domain corpora, models spanning various architectures possess unparalleled world knowledge, natural language understanding, and complex reasoning capabilities[7]. The integration of these foundational models into recommender systems has birthed the paradigm of generative recommendation, which eliminates the traditional multi-stage filtering pipeline in favor of a single-stage, end-to-end generation process[4]. Instead of computing discrete relevance scores for predefined candidate pools and relying on approximate nearest neighbor search, generative recommenders autoregressively decode the exact identifier of the target item conditioned on the user's context and historical interaction sequences[15].

Embracing generative recommendation unlocks multiple intrinsic advantages. The integration of vast world knowledge allows systems to naturally incorporate complex item metadata—such as cultural trends, brand relationships, and descriptive semantics—without requiring explicitly constructed knowledge graphs or auxiliary feature pipelines[3]. The advanced natural language understanding capabilities of LLMs bridge the gap between implicit behavioral logs and explicit human intent, enabling interactive, conversational recommendation interfaces[3]. Moreover, the autoregressive formulation provides a unified mathematical framework for previously disparate tasks, converging retrieval, ranking, explanation generation, and cross-domain transfer into a singular conditional sequence generation objective[20].

Despite these profound theoretical advantages, the deployment of LLM-based generative recommendation introduces significant representational and computational complexities. Standard natural language tokens are highly suboptimal for representing structured, sparse, and massive item catalogs. Furthermore, LLMs pre-trained exclusively on natural language often lack the inductive biases necessary to capture the high-order collaborative signals embedded within user-item interaction graphs[23]. To provide a comprehensive synthesis of this rapidly evolving domain, this survey systematically categorizes recent advancements into focused methodological directions, spanning from foundational tokenization strategies and collaborative knowledge fusion to lifelong user behavior modeling and alignment via reinforcement learning.

Methods

The integration of LLMs into the recommendation domain can be classified into distinct methodological families based on data representation, structural adaptation, and generative task formulation. The following subsections systematically decompose these core architectures and the innovations driving generative recommendation.

Recommendation as Language Processing (RLP)

The earliest and most fundamental approach to leveraging LLMs for recommendation involves transforming structured tabular data, user-item interactions, and item metadata into unified natural language sequences[20]. This approach, formally conceptualized as Recommendation as Language Processing (RLP), utilizes instruction tuning to teach LLMs to perform recommendation tasks via pure text generation.

A foundational framework in this domain is the Pretrain, Personalized Prompt, and Predict Paradigm (P5)[20]. P5 converts varied recommendation tasks—including sequential recommendation, direct item recommendation, explanation generation, and review summarization—into a shared text-to-text format. By optimizing a unified negative log-likelihood objective over these diverse prompt-response pairs, P5 allows a single LLM to serve as a universal foundation model for the entire recommendation ecosystem. The model executes zero-shot and few-shot recommendations by processing personalized prompts and autoregressively generating textual responses, thereby eliminating the need for task-specific architectural modifications[20].

Industrial-scale implementations have further demonstrated the viability of open-ended task unification using generative pre-trained transformers. The M6-Rec framework, built upon a massive industrial LLM, addresses strict hardware constraints by replacing full-parameter fine-tuning with highly efficient prompt tuning[22]. By employing late interaction, parameter sharing, and early-exiting mechanisms, M6-Rec compresses the model size and accelerates inference, allowing the deployment of generative models on both cloud servers and mobile edge devices while maintaining cross-domain transferability[22].

To overcome the discrepancy between the general pre-training objectives of LLMs and specific recommendation tasks, methodologies like TALLRec propose lightweight instruction tuning frameworks[27]. By utilizing Low-Rank Adaptation (LoRA) on architectures such as LLaMA, TALLRec demonstrates that LLMs can be effectively aligned with recommendation logic using fewer than one hundred carefully curated tuning samples, facilitating robust cross-domain generalization (e.g., from movie to book recommendations)[27]. The ExplainRec extension further augments this approach by introducing preference attribution tuning, formatting the instruction data to explicitly require the model to output a natural language rationale alongside the binary prediction, thereby resolving the black-box nature of traditional recommendations[30].

RLP Paradigm	Core Mechanism	Primary Advantages	Structural Limitations
P5 Framework [cite: 20, 21]	Unified text-to-text conversion of all tasks via personalized prompts.	Universal task application; strong zero-shot capabilities.	High inference latency; context window explosion.

M6-Rec [cite: 22, 29]	Parameter-efficient prompt tuning with late interaction and early exiting.	Edge-device deployability; reduced parameter updates (1%).	Requires complex engineering to maintain throughput.
TALLRec [cite: 27, 31]	LoRA-based instruction tuning on specialized recommendation corpora.	Highly sample-efficient; robust cross-domain transfer.	Relies heavily on high-quality textual metadata.

Despite these successes, representing items exclusively through natural language titles or descriptions creates critical bottlenecks. Textual identifiers suffer from severe semantic ambiguity, where token overlap across unrelated items confuses the generative decoder[15]. Additionally, natural language sequences expand the context length significantly, resulting in prohibitive computational costs during multi-head self-attention, rendering pure RLP impractical for industrial candidate retrieval[33].

Generative Retrieval and Semantic Tokenization

To overcome the inefficiencies and semantic ambiguities of natural language item representations, the field has aggressively pivoted toward Semantic Identifiers (SIDs). In this generative retrieval paradigm, items are mapped to discrete, learnable tokens that capture semantic and collaborative topologies, allowing the LLM to output a precise, compact identifier directly from the entire corpus[15].

Residual-Quantized Variational AutoEncoders (RQ-VAE)

The Transformer Index for Generative Recommenders (TIGER) introduced a breakthrough methodology for indexing items via hierarchical quantization[16]. Instead of matching dense embeddings via approximate nearest neighbor search, TIGER trains a sequence-to-sequence model to directly decode Semantic IDs. To generate these IDs, an item's textual and metadata features are first passed through a dense text encoder. The resulting continuous embedding is discretized using a Residual-Quantized Variational AutoEncoder (RQ-VAE)[16].

The RQ-VAE operates over a predefined number of hierarchical levels, m. At the first level, it assigns the item to the nearest cluster centroid in a learned codebook. At subsequent levels, it quantizes the residual error from the previous step. This algorithm maps every item to an m-tuple of discrete tokens: $(c_0, c_1, \ldots, c_{m-1})$. The generation probability of an item is therefore factorized autoregressively:

$$P(\mathrm{ID}) = \prod_{k=0}^{m-1} P(c_k \mid c_0, \ldots, c_{k-1}, \mathrm{history})$$

This formulation creates a hierarchical taxonomy where similar items share overlapping prefix tokens, fundamentally embedding semantic similarity into the identifier's syntactic structure and dramatically shrinking the search space[16].

Mitigating Hard Quantization Bottlenecks

While RQ-VAE SIDs reduce context length and memory footprints, they suffer from hard residual quantization, which forcibly collapses multi-faceted item semantics at arbitrary cluster boundaries and propagates early quantization errors to later SID positions[37].

To address the dual-stage information bottleneck, the AsymRec framework introduces an asymmetric continuous-discrete architecture that decouples the input and output representations[38]. It introduces Multi-expert Semantic Projection (MSP) to map continuous embeddings directly into the Transformer's hidden space for the input context, circumventing lossy quantization during encoding. Simultaneously, it utilizes Multi-faceted Hierarchical Quantization (MHQ) to construct high-capacity discrete targets for the output, preventing dimensional collapse while retaining fine-grained distinctions[38].

An alternative innovation is CapsID, which alters the fundamental assignment operator by replacing hard nearest-neighbor quantization with soft capsule routing[37]. At each semantic layer, an item probabilistically routes to several semantic capsules. The residual vector is updated by the routed reconstruction rather than a single winning code, and the tokenization process terminates dynamically once an active capsule reaches a predefined confidence threshold[37]. This soft-routed, variable-length tokenization preserves boundary semantics, providing significant recall improvements, particularly for long-tail items that are historically marginalized by rigid clustering[39].

Bridging Generative and Dense Paradigms

Generative retrieval using discrete SIDs excels in memory efficiency and cold-start generalization but occasionally underperforms dense retrieval models that leverage continuous, high-dimensional spaces for fine-grained ranking[36]. Dense retrieval computes inner products between continuous user and item representations, optimizing for exact-match precision but scaling poorly with massive candidate sets[36].

Hybrid approaches attempt to fuse the scalability of generative retrieval with the precision of dense retrieval. The LIGER (LeveragIng dense retrieval for GEnerative Retrieval) framework utilizes generative SIDs to retrieve a highly truncated, coarse candidate set, followed by a sequential dense retrieval module for precision ranking. This mitigates performance discrepancies and enhances efficiency on cold-start items while retaining a minimal storage footprint[36].

The COBRA (Cascaded Organized Bi-Represented generAtive retrieval) framework unifies this dynamic entirely within a single end-to-end Transformer decoder[39]. COBRA models the item identifier as a cascaded sequence consisting of a sparse Semantic ID followed by a dense continuous vector. During end-to-end training, the dense representations are refined dynamically via contrastive learning objectives, capturing deep collaborative signals. During inference, COBRA employs a coarse-to-fine generation strategy: the model first autoregressively generates the sparse semantic ID, which provides a high-level categorical sketch. The generated ID is then appended to the context and fed back into the model to condition the generation of the fine-grained dense vector[43]. To ensure flexible inference, COBRA introduces BeamFusion, a sampling technique combining beam search probabilities with nearest neighbor retrieval scores, ensuring highly controllable diversity in the retrieved items[42].

Tokenization Strategy	**Representation Format**	**Core Mechanism**	**Inference Characteristics**
RQ-VAE SIDs [cite: 16]	Discrete m -tuple tokens	Hard nearest-neighbor residual quantization.	Autoregressive beam search; high memory efficiency.
AsymRec [cite: 38]	Asymmetric Continuous-Discrete	MSP for input embeddings; MHQ for output tokens.	Preserves input richness; avoids quantization error on context.
CapsID [cite: 39]	Variable-length discrete tokens	Probabilistic soft capsule routing with confidence thresholds.	Dynamic length decoding; highly effective for long-tail items.
COBRA [cite: 43]	Cascaded Sparse + Dense	End-to-end optimization of IDs and continuous vectors.	Coarse-to-fine generation via BeamFusion.

Fusing Collaborative Signals with LLM Semantic Spaces

A significant and documented limitation of off-the-shelf LLMs is their "collaborative blindness." While they excel at reasoning over raw text and extracting semantic item attributes, they inherently lack the topological awareness of user-item interaction graphs and the high-order collaborative filtering (CF) signals that traditional recommender systems rely upon[23]. Merely injecting raw numeric item IDs into an LLM severely degrades performance, as numeric IDs lack semantic grounding within the LLM's pre-trained vocabulary and fail to trigger its internal reasoning circuits[1].

To reconcile this discrepancy, researchers have developed sophisticated mechanisms to inject external collaborative knowledge directly into the LLM's representation space, treating collaborative structure as an auxiliary modality[46].

Direct Modality Projection

The CoLLM framework treats collaborative information as an entirely distinct modality[48]. CoLLM extracts user and item embeddings from an external, well-trained CF model (e.g., LightGCN or matrix factorization) and trains a dedicated mapping network to project these dense CF embeddings directly into the LLM's input token space[48]. This approach preserves the LLM's frozen semantic reasoning capabilities while explicitly anchoring its predictions in empirical interaction structures. The framework employs a two-step tuning procedure: initial fine-tuning using pure language information, followed by specialized tuning of the mapping module to ensure the mapped collaborative data is interpretable by the LLM[49].

Curriculum Learning and Hybrid Prompting

The LLaRA (Large Language-Recommendation Assistant) model adopts a similar projector design but introduces a progressive curriculum learning strategy[51]. Recognizing that exposing an LLM immediately to unfamiliar continuous CF embeddings causes representational collapse, LLaRA utilizes a hybrid prompt design. The model is initially warmed up on pure text-based prompts, characterizing items through textual metadata, which aligns seamlessly with the LLM's natural language modeling objective. As training progresses, the system iteratively shifts to hybrid prompts that replace specific text tokens with the projected ID embeddings, smoothly and stably transitioning the LLM from semantic matching to complex collaborative behavioral reasoning[51].

Semantic Alignment and Attention Steering

The SeLLa-Rec framework notes that purely mathematical projection is insufficient if the CF knowledge distribution severely mismatches the LLM's pre-trained latent space[45]. SeLLa-Rec focuses on Semantic Alignment, integrating collaborative knowledge through item-side distillation and contrastive alignment before injection. Building upon these diagnostic insights, SAILRec (Steering Attention to collaborative embeddings In LLM Recommendation) introduces a hierarchical attention steering mechanism[55]. Extensive analyses reveal that LLMs utilize injected collaborative embeddings in a highly depth-dependent manner[56]. SAILRec actively suppresses collaborative interference in the shallow Transformer layers (allowing the model to perform foundational semantic parsing of the context) and dynamically amplifies collaborative attention in the deeper decision layers, optimizing the integration of CF constraints and semantic priors for final preference matching[55].

Similarly, the RecMind framework treats the LLM as a preference prior rather than an end-to-end monolithic ranker. RecMind utilizes lightweight adapters to condition a frozen LLM on textual metadata, while a Graph Neural Network (GNN) simultaneously learns collaborative embeddings from the interaction graph. These two distinct views are aligned through a symmetric contrastive objective and fused via an intra-layer gating mechanism, allowing language semantics to dominate in cold-start regimes and graph structures to stabilize rankings in data-rich scenarios[24].

User Lifelong Behavior Modeling (ULBM)

Industrial recommender systems must track users across highly protracted time horizons, leading to the computational challenge of User Lifelong Behavior Modeling (ULBM). Users accumulate tens of thousands of heterogeneous interactions (e.g., clicks, views, additions to cart, purchases), creating sequences that easily exceed the maximum context windows of standard Transformers[10]. Furthermore, standard full-attention mechanisms scale quadratically with sequence length, $\mathcal{O}(L^2)$, rendering them computationally intractable for real-time lifecycle modeling and prone to stochastic noise accumulation[59].

Recurrent Memory Architectures

To bypass the quadratic context limitations of standard Transformers, generative recommenders have increasingly adopted memory-augmented recurrent architectures. The Rec2PM (Recurrent Preference Memory) framework compresses ultra-long user interaction histories into compact, latent "Preference Memory" tokens[61]. Instead of maintaining massive Key-Value (KV) caches, Rec2PM continuously updates this tokenized memory state. To circumvent the slow, serial optimization typical of Backpropagation Through Time (BPTT), Rec2PM employs a novel self-referential teacher-forcing strategy. It utilizes a global, full-attention view of the entire sequence during offline training to generate ideal "reference memories," which subsequently serve as parallel supervision targets for the recurrent updates[61]. This allows fully parallelized offline training while supporting highly efficient $\mathcal{O}(1)$ iterative state updates during online inference, effectively acting as a denoising information bottleneck that filters out stochastic noise while capturing robust long-term interests[61].

Multi-Stream and Selective Segmentation

Other approaches handle massive sequences via temporal routing and sparse attention. The GEMs (Generative rEcommendation with a Multi-stream decoder) framework treats long-sequence modeling as a multi-timescale memory reading problem, partitioning user histories into three distinct temporal streams: Recent, Mid-term, and Lifecycle[63]. It processes recent behaviors with standard dense attention, utilizes a lightweight cross-attention indexer to extract relevant signals from the massive mid-term stream, and leverages a parameter-free fusion mechanism to integrate macro-level lifecycle trends, thoroughly breaking the long-sequence barrier[63].

Similarly, the GRACE framework leverages a Journey-Aware Sparse Attention (JSA) mechanism applied over Chain-of-Thought (CoT) tokenization. Rather than attending to all historical tokens, JSA selectively attends only to highly relevant intra-sequence contexts and explicitly reasoned attributes from product knowledge graphs (e.g., category, brand, price). This mechanism ensures interpretable generation while reducing quadratic computational overhead by up to 48% on extended sequences[60].

Synthetic Behavior Generation and Simulation

Addressing the inherent sparsity of long-term tracking data and the strict privacy constraints surrounding user behavior logs, LLMs are increasingly being deployed as generative simulators to synthesize highly realistic behavioral trajectories[64]. The BehaveGPT foundation model is pre-trained on hundreds of millions of user behavior logs using a Distributionally Robust Optimization (DRO) paradigm. This specific pre-training paradigm ensures that the model captures complex behavior patterns equitably, mitigating the popularity bias that causes models to overfit head items while ignoring tail behaviors[65].

Concurrently, the BehaviorGen framework utilizes psychological profiles and a sparse set of real, localized historical events to prompt LLMs into generating high-fidelity synthetic human mobility and digital usage sequences[64]. By carefully balancing population-level diversity with individual-level specificity, BehaviorGen serves as a robust, privacy-preserving data augmentation strategy. When applied as a fine-tuning replacement or augmentation tool for downstream discriminative models, it has demonstrated substantial predictive gains across empirical studies[64].

Conversational and Agentic Recommendation Systems

The generative nature of LLMs inherently supports continuous, multi-turn dialogue, fundamentally facilitating the evolution of Conversational Recommender Systems (CRSs). Unlike traditional recommenders that infer latent intent solely from passive clicks, CRSs actively elicit user preferences through natural language inquiry, allowing for the real-time clarification of ambiguous or shifting needs[68].

LLM-Augmented Conversational Interfaces

Systems such as Chat-Rec leverage LLMs as an interactive conversational front-end, continuously refining prompt-based queries based on user dialogue[19]. The framework operates by compressing massive candidate sets—originally retrieved by traditional dense backend models—into concise natural language contexts. This allows the LLM to seamlessly weave explicit rationale, comparisons, and explanations into its conversational responses, drastically improving perceived system transparency and user trust. Moreover, prompt-based injection allows the conversational agent to handle cold-start items efficiently by contextualizing new item features dynamically without requiring complete model retraining[19].

Autonomous Recommendation Agents

Moving beyond reactive, single-turn chatbots, architectures such as RecMind deploy the LLM as an autonomous agent capable of complex, multi-step planning, logical reasoning, and external tool usage[57]. RecMind introduces a "Self-Inspiring" planning algorithm. At each intermediate reasoning step, the LLM evaluates all previously explored behavioral states and logical pathways, explicitly deciding whether to exploit the current path or explore alternative query strategies[73]. This allows the agent to dynamically query external databases, verify structural constraints (e.g., calculating budget limitations or checking inventory), and construct highly personalized zero-shot recommendations firmly grounded in factual external knowledge[73].

Multi-Agent Coordination

Addressing the extreme difficulty of balancing conflicting optimization objectives (e.g., maintaining accuracy, diversity, novelty, and conversational coherence simultaneously) within a single set of model weights, the AgentRec framework introduces a distributed multi-agent topology[70]. It distributes specific recommendation tasks across specialized, parallel LLM agents—comprising a Conversation Understanding Agent, a Preference Modeling Agent, a Context Awareness Agent, and a Dynamic Ranking Agent. These nodes operate collaboratively via an adaptive coordination mechanism, dynamically learning optimal influence weights based on the complexity and semantic depth of the ongoing conversational context, significantly reducing user abandonment rates during complex multi-criteria decision scenarios[70].

Multi-Level Alignment and Reinforcement Learning

As generative recommenders transition from simple supervised next-token prediction (NTP) to optimizing long-term user satisfaction and multi-turn engagement, Reinforcement Learning (RL) and preference alignment techniques have become critical infrastructural components[15]. Traditional Supervised Fine-Tuning (SFT) often suffers from exposure bias and cannot effectively capture implicit, relative user satisfaction metrics derived from real-world engagement (e.g., varying dwell times, partial clicks)[35].

To systematically address behavioral and semantic misalignment, the Align3GR framework unifies alignment across three architectural levels: token-level (via dual semantic-collaborative IDs), behavior modeling-level (via multi-task SFT), and preference-level (via Direct Preference Optimization, DPO)[76]. Align3GR utilizes a progressive DPO strategy—incorporating specific scenario preference and real-feedback DPO—that meticulously organizes training data from easy to hard preference pairs. This progressively stabilizes the alignment of the LLM's generative probability distribution with actual recommendation utilities, yielding massive improvements in offline ranking metrics and real-world click-through rates[76].

For robust industrial-scale optimization, the GenRec framework implements Group Relative Policy Optimization with Supervised Regularization (GRPO-SR)[35]. Traditional reinforcement learning in generative models frequently suffers from reward hacking, where the policy learns to output syntactically valid but semantically useless token combinations that manipulate the reward model[75]. GenRec mitigates this via a stringent hybrid gating mechanism $r = G \cdot r'$, where the gate $G = \mathbb{I}(s > \tau)$ outright suppresses irrelevant or hallucinated trajectories. Furthermore, GRPO-SR explicitly anchors relative preference scores within a grouped generative rollout, optimizing the advantage function A against a supervised cross-entropy baseline to guarantee stable, single-step policy updates across highly sparse, large-scale feedback loops[75].

Challenges and Prospects

While the theoretical efficacy and representational power of LLM-based generative recommendation are well established, scaling these architectures to meet the rigorous constraints of real-world industrial deployments presents persistent and multifaceted challenges.

Inference Latency and Industrial Scalability

The fundamental operational bottleneck of generative recommendation is the high inference latency inherent to autoregressive decoding, which directly conflicts with the strict millisecond Service Level Agreements (SLAs) required by real-time platforms[46]. Multi-step beam search, absolutely required to decode complex, multi-token Semantic IDs accurately, drastically inflates the necessary FLOPs per user request.

To bridge this massive efficiency gap, architectures such as RankGPT transition the generative paradigm back toward highly optimized ranking operations, demonstrating that the reasoning power of LLMs can be utilized effectively within constrained discriminative architectures[80]. Additionally, comprehensive system-level serving optimizations have been introduced in frameworks like GR4AD. These include Dynamic Beam Serving (DBS), which adaptively adjusts the beam width sequence based on the model's confidence, and Traffic-Aware Adaptive Beam Search (TABS), which modulates the overall beam scale according to instantaneous server traffic and available computational slack[79]. Furthermore, GR4AD implements Beam-Shared KV Caching, organizing beams along the sequence dimension so that multiple decoding paths share a single encoder KV cache. This critical optimization eliminates redundant memory accesses, reducing the per-step KV read complexity from $\mathcal{O}(B \cdot L)$ to $\mathcal{O}(L)$, ensuring that generative recommenders can function under peak commercial loads[79].

Furthermore, generative retrieval is highly sensitive to model capacity and data volume. Empirical studies investigating neural scaling laws in generative retrieval demonstrate that increasing inference-time compute yields predictable, power-law performance gains. These studies reveal that large-scale decoder-only architectures (e.g., LLaMA) consistently exhibit higher theoretical upper bounds and greater efficiency than encoder-decoder models (e.g., T5) when modeling complex item indices, underscoring the critical interplay between architecture selection, dataset scale, and computational budgets[82].

Temporal Generalization and Evaluation Methodologies

Evaluating the true reasoning capabilities of generative recommenders requires moving decisively beyond static, single-domain next-item prediction benchmarks. The recently introduced HORIZON benchmark highlights that modern generative user models struggle significantly with out-of-distribution (OOD) generalization, specifically when forced to adapt to entirely unseen users or when extending predictions across highly protracted temporal horizons[85]. Future research must prioritize robust inductive settings, where models are forced to recommend entirely novel, cold-start items dynamically without retraining the index. Frameworks applying speculative decoding and "draft-verify" techniques show immense promise in enabling these zero-shot inductive capabilities, allowing the generative framework to verify new items against dense attributes on the fly[86].

Additionally, deploying generative models in complex multi-business ecosystems (e.g., super-apps simultaneously offering food delivery, ride-hailing, and retail shopping) creates severe cross-domain signal interference. Frameworks like Multi-Business Generative Recommendation (MBGR) attempt to disentangle these conflicting signals using domain-aware semantic tokenization (Business-aware IDs) and label dynamic routing, ensuring knowledge sharing across domains while preserving business-specific specificity. However, managing the gradient conflicts between disparate business objectives during multi-task SFT remains a highly active area of investigation[87].

Machine Unlearning and Privacy Retention

As generative models ingest and memorize vast quantities of historical interactions directly within their parametric weights, they pose severe data privacy risks. If a user exercises their right to be forgotten and requests the deletion of their behavioral data, retraining massive LLMs from scratch is computationally intractable and economically unviable. Machine unlearning in LLM-based recommenders is heavily confounded by the "Polysemy Dilemma," wherein the specific neuronal pathways that encode sensitive privacy data simultaneously overlap with the pathways required for general logical reasoning[88].

The U-CAN (Utility-aware Contrastive Attenuation) framework directly addresses this dilemma by applying adaptive soft attenuation. By analyzing contrastive activation differences between a targeted forgetting set and a general retention set, U-CAN isolates asymmetric neuronal responses. Instead of utilizing rigid binary pruning that destroys network connectivity, U-CAN dampens high-risk parameters located on Low-Rank Adapters (LoRA) using a precisely calibrated, differentiable decay function. This mechanism successfully suppresses sensitive retrieval pathways and guarantees privacy compliance without catastrophically destroying the model's structural reasoning capabilities[88].

Algorithmic Fairness and Bias Mitigation

Algorithmic fairness in LLM-augmented recommenders requires urgent and systematic attention. Biases inherited from massive pre-training corpora can inadvertently skew recommendations toward historically popular items, specific demographic aesthetics, or overrepresented societal norms[72]. Current literature emphasizes the critical need for fairness-aware decoding strategies, prompt debiasing, and explicitly generated rationales to ensure item-side and user-side equity[90]. Models like ExplainRec leverage preference attribution tuning and chain-of-thought (CoT) generation to expose the logical grounding behind recommendations. By forcing the LLM to articulate exactly why an item was chosen, these systems enable transparent, third-party audits of algorithmic equity, mitigating implicit bias and fostering long-term user trust[30].

Conclusion

The integration of Large Language Models into recommender systems signifies a permanent and necessary departure from isolated, discriminative filtering algorithms. By reformulating the recommendation problem as an autoregressive conditional sequence generation task, the field has unlocked the unprecedented ability to unify highly fragmented data modalities, leverage deep semantic reasoning, and interact dynamically with users through sophisticated conversational agents. Critical architectural breakthroughs—including Residual-Quantized Semantic IDs, collaborative-semantic feature alignment, and recurrent preference memory—have successfully mitigated the bottlenecks of deploying natural language processing models on sparse, mathematically rigid behavioral graphs.

However, realizing the full potential of these generative architectures requires sustained, multi-disciplinary innovation in efficient inference mechanisms, sophisticated reinforcement learning alignment, robust inductive generalization, and rigorous privacy controls. As neural scaling laws continue to drive raw model capabilities upward, the ongoing synthesis of natural language understanding and topological graph reasoning will inevitably give rise to next-generation, autonomous recommendation agents capable of holistic, transparent, and seamlessly personalized user lifecycle management across diverse digital ecosystems.

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